

twist-tunable stacks of other types of 2D materials. It turns out moiré effects can be obscured somewhat by an imperfect moiré pattern, which will lead to light emission similar to defects, and excess laser excitation power, which will provide a broad background.

The science and engineering behind twistronics

Achieving a variety of electronic properties in conventional materials normally requires changing their chemical composition by doping technique. Researchers in recent times have realised that weak coupling between different layers of 2D materials can be used to control these materials in ways that were not possible with more conventional structures.

One unique example in this direction is the ability to tune electronic properties of 2D-layered graphene by varying angle between the layers by twisting. Thus, ability to vary electronic properties of a 2D material just by altering the twist angle between its layers is therefore a fundamentally new direction in device engineering.

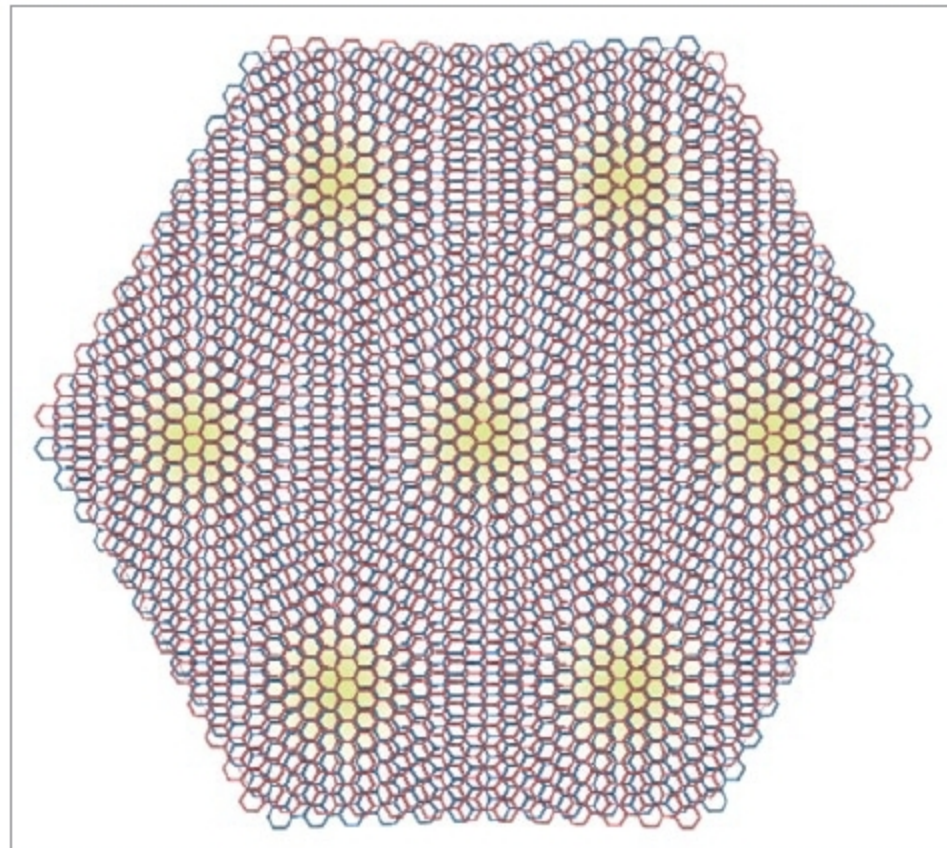
Until now, researchers needed to make a whole new device for each different twist angle, which often involved fabricating a large number of devices. This also made it difficult to resolve angle-dependent effects independent of sample-to-sample variations. Many interesting angle-dependent characteristics (such as superconductivity in twisted bilayer graphene) could only be observed within a very narrow range of the target angle.

Researchers have now succeeded in making a single device whose layers can continuously rotate locally while measuring its physical and electronic properties. This new platform allows switching between an arbitrary number of complementary states in the device.

Researchers at Columbia University, USA, have developed a new device structure to make it possible to

vary the twist angle between layers of 2D materials and to investigate how this angle affects electronic, optical and mechanical properties of the material. Measurements carried out on a single structure with twisting rather than multiple ones—as was the case before—could advance the emerging field of twistronics as a fundamentally new approach to device engineering.

For instance, graphene, a 2D sheet of carbon atoms, normally does not have a band gap, but it develops one when placed in contact with another



Magic angle graphene superlattice; scale=10nm
(Credit: <https://physicsworld.com>)

2D material, boron-nitride, with a hexagonal structure. Both these materials have closely matching lattice constants, and layers of graphene and boron-nitride form a large moiré superlattice. Then, by twisting the layers so that they become misaligned and angle between them becomes large, the band gap disappears. Simply varying the angle between 2D material layers, thus, means that graphene can be tuned from metallic to semiconducting.

Researchers at Massachusetts Institute of Technology (MIT), USA, have recently discovered that placing two layers of graphene together, when rotated relative to one another at the magic angle of 1.1, turns the normally metallic material into a superconductor.

Basics of the technique

This technique takes advantage of low interfacial friction that naturally exists between layered 2D materials, and also absence of strong chemical binding between the layers so they slide over one another easily. In the fabrication process, researchers studied graphene/boron-nitride heterostructures and designed a device in the shape of a gear that was intentionally made so that it was rotatable. Previously-developed techniques were employed to mechanically pick up this nano-sized gear (made of boron-nitride) and place it over the top of the active area (graphene).

Once in place, they used an atomic force microscope to push along one tooth of the gear, causing it to rotate. This shows that we can achieve and control rotation in graphene/boron-nitride heterostructures, and can dynamically vary electrical, optical and even mechanical properties of a device made from these structures. It is, thus, demonstrated that energy gap observed in graphene is tunable and can be turned on or off as per requirement just by changing the

twist angle between the layers. Further spectroscopic studies of twisted bilayer graphene have revealed strong electron-electron correlations at the magic angle.

It is already known that graphene can conduct electricity at ultra-high speed, but researchers took a giant leap by turning graphene into a superconductor—a material that allows electricity to flow without resistance. This was achieved by placing one sheet of graphene over another and rotating the other sheet to a special orientation, or magic angle, and cooling the ensemble to a temperature of 1.7K. The twist radically changed the bilayer's properties—turning it first into an insulator and then, with application of a stronger electric field, into a superconductor.